

# PAPER\_890: SCm Vacuum Density Evolution

**Author:** Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** SCmVacuumDensityEvolutionCalc (CP4 #474) **CVW:** v2.0.0 compliant

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## Abstract

Time evolution of the SCm vacuum density  $\rho_{\text{SCm}}(t) = \rho_{\text{vac,SCm}} \cdot S_{26}([SSq]) \cdot \exp(\kappa t + [SSq]t/26)$ . The SCm/UA density ratio  $\rho_{\text{SCm}}/\rho_{\text{UA}} = 0.1$  provides the hierarchy between superconductive matter and undifferentiated aether, with Hubble-normalized evolution.

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## 1. Core Equations

$$\begin{aligned}\rho_{\text{SCm}}(t) &= \rho_{\text{vac,SCm}} \cdot S_{26} \cdot \exp(\kappa t + [SSq]t/26) \\ \rho_{\text{SCm}}/\rho_{\text{UA}} &= 0.1\end{aligned}$$

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## 2. Parameters

| Parameter   | Default                    | Description        |
|-------------|----------------------------|--------------------|
| t           | 0.0 s                      | Time parameter     |
| rho_vac_scm | 9.47e-27 kg/m <sup>3</sup> | SCm vacuum density |

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## 3. UQFF Integration

This calculator integrates `et_scm_vacuum.py` from Session 207 into the CP4 pipeline as class #474.

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